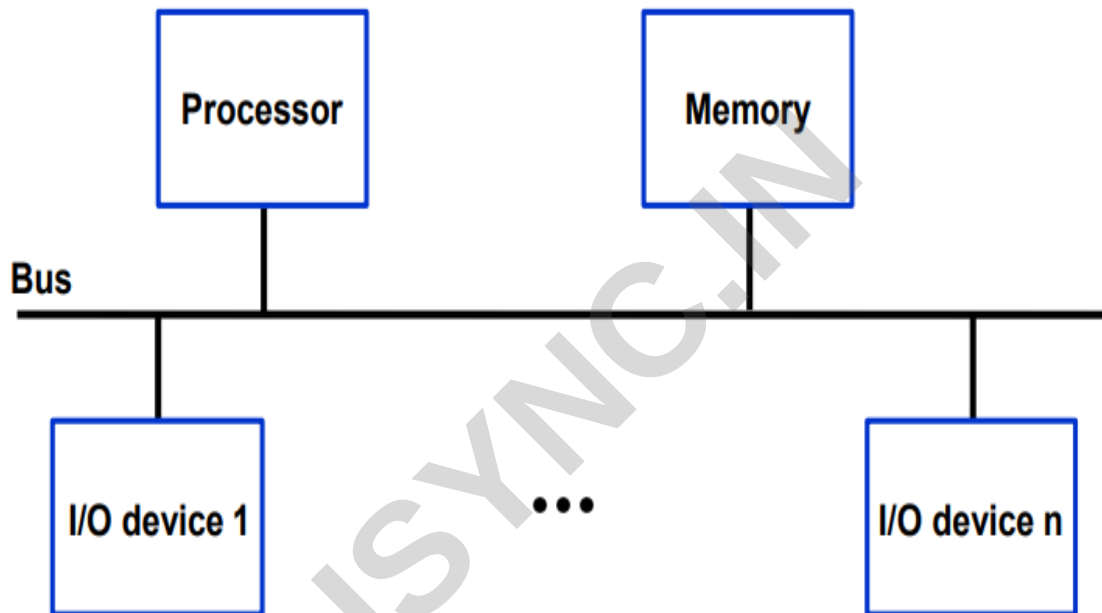


Module-3 Input and Output Organization

Accessing I/O-Devices:

A **single bus-structure** can be used for connecting I/O-devices to a computer. The simple arrangement of connecting set of I/O devices to memory and processor by means of system bus is as shown in the figure. Such an arrangement is called as Single Bus Organization.



- The single bus organization consists of
 - Memory
 - Processor
 - System bus
 - I/O device
- The system bus consists of 3 types of buses:
 - Address bus (Unidirectional)
 - Data bus (Bidirectional)
 - Control bus (Bidirectional)
- The system bus enables all the devices connected to it to involve in the data transfer operation.
- The system bus establishes data communication between I/O device and processor.

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- Each I/O device is assigned a unique set of address.
- When processor places an address on address-lines, the intended-device responds to the command.
- The processor requests either a read or write-operation.
- The requested-data are transferred over the data-lines



Steps for input operation:

- The address bus of system bus holds the address of the input device.
- The control unit of CPU generates IORD Control signal.
- When this control signal is activated the processor reads the data from the input device(DATAIN) into the CPU register.

Steps for output operation:

- The address bus of system bus holds the address of the output device.
- The control unit of CPU generates IOWR control signal.
- When this control signal is enabled CPU transfers the data from processor register to outputdevice(DATAOUT)



There are 2 schemes available to connect I/O devices to CPU

1. Memory mapped I/O:

- In this technique both memory and I/O devices can share the common memory to store the instruction as well as the operands.
- Memory related instructions are used for data transfer between I/O and processor.
- In case of memory mapped I/O input operation can be implemented as,

MOVE DATAIN, R0
- This instruction sends the contents of location DATAIN to register R0. Similarly output can be implemented as,

MOVE R0, DATAOUT

2. I/O Mapped I/O:

- In this technique CPU separates address space for memory and I/O devices.

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- Hence two sets of instruction are used for data transfer.
- One set for memory operations and another set for I/O operations.
- The I/O operation can be implemented as, IN AL, DX
- This instruction reads one byte of data from the I/P register whose address is store in DX register into AL register.
- The O/P operation can be implemented as, OUT DX, AL
- This instruction transfer the contents of AL register into the O/P register (DATA OUT) whose address is stored in DX register.

Comparison between I/O mapped I/O and Memory mapped I/O:

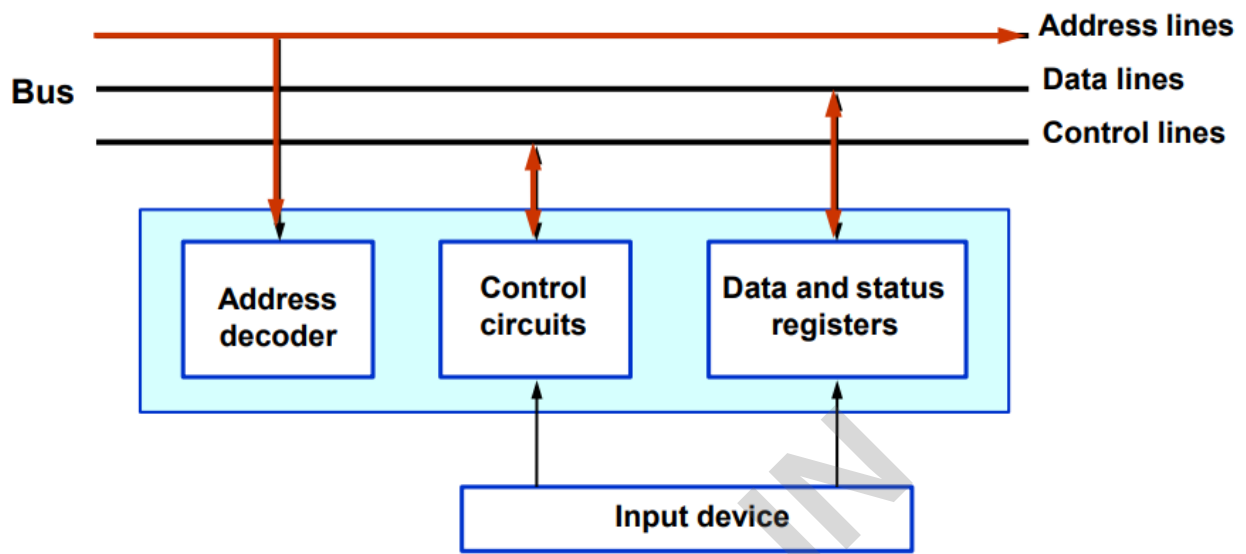
Isolated I/O	Memory-mapped I/O
Different address spaces are used for computer memory and I/O devices. I/O devices have dedicated address space.	Same address space is used for memory and I/O devices.
Separate control unit and control instructions are used in case of I/O devices.	Control units and instructions are same for memory and I/O devices.
More complex and costlier than memory-mapped I/O as more bus are used.	Easier to build and cheap as it's less complex.
Entire address space can be used by memory as I/O devices have separate address space.	Some part of the address space of computer memory is consumed by I/O devices as address space is shared.
Computer memory and I/O devices use different control instructions for read write.	Computer memory and I/O devices can both use same set of read and write instructions.
Separate control bus is used for computer memory and I/O devices. Though same address and data bus are used.	Address, data and control bus are same for memory and I/O devices.



I/O Interface for an Input Device:

The hardware arrangement of connecting I/P device to the system bus is as shown in the fig.

Module-3 Input and Output Organization



This hardware arrangement is called as I/O interface. The I/O interface consists of 3 functional devices namely:

1) Address Decoder:

- Its function is to decode the address in-order to recognize the input device whose address is available on the unidirectional address bus.
- The unidirectional address bus of system bus is connected to input of the address decoder as shown in figure

2) Control Circuit:

- The control bus of system bus is connected to control circuit as shown in the fig.
- It controls the read write operations with respect to I/O device.

3) Status & Data register:

- It specifies type of operation (either read or write operation) to be performed on I/O device.

4) Data Register:

It stores the data to be read from input device to or it holds the data to be written into output device.

There are 2 types:

DATAIN - Input-buffer associated with keyboard.

DATAOUT -Output data buffer of a display/printer.



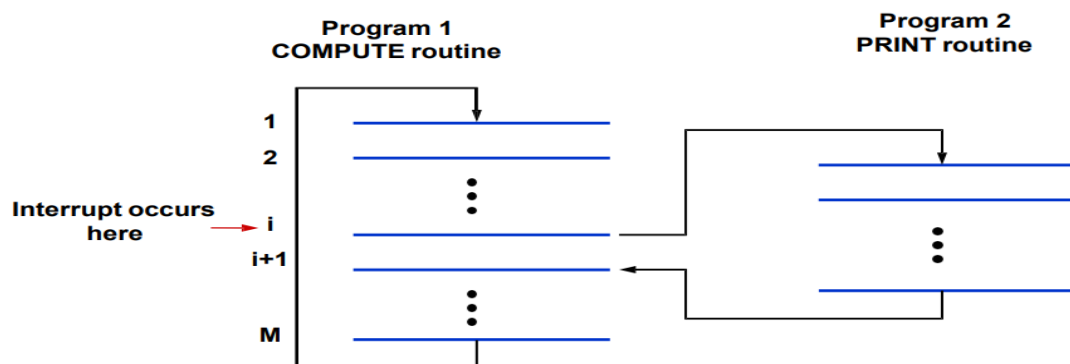
Module-3 Input and Output Organization

Program Controlled I/O:

- To explain the concept of program controlled I/O consider two operations namely input operation and output operation.
 - a) Input operation:
 - It is the process of transferring ASCII value of a character from DATA IN register to CPU register.
 - b) Output operation:
 - It is the process of transferring ASCII value of a character from CPU register to DATA OUT register.
- Design 2 sets of instruction for input and output operations respectively,
- These 2 sets of instructions are stored in memory of the CPU. Now allow the CPU to execute this program in-order to control the input and output operation.
- “It is the process of controlling the input and output operations by executing 2 sets of instruction, one set for input operation and the next set for output operation.”

Interrupt:

- It is an event which suspends the execution of one program and begins the execution of another program.
- The arrival of interrupt causes the processor to transfer the execution control from main program to sub program.
- To explain the concept of interrupt, consider the following two programs namely:



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The following steps take place when the interrupt related instruction is executed:

- It suspends the execution of current instruction i.
- Transfer the execution control to sub program from main program.
- Increments the content of PC by 4 memory location.
- It decrements SP by 4 memory locations.
- Pushes the contents of PC into the stack segment memory whose address is stored in SP.
- It loads PC with the address of the first instruction of the sub program.

The following steps takes place when return instruction is executed

- It transfers the execution control from sub program to main program.
- It retrieves the content of stack memory location whose address is stored in SP into the PC.
- After retrieving the return address from stack memory location into the PC it increments theContent of SP by 4 memory location.

Interrupt Latency is a delay between

- time an interrupt-request is received and
- start of the execution of the ISR.

- Generally, the long interrupt latency in unacceptable.

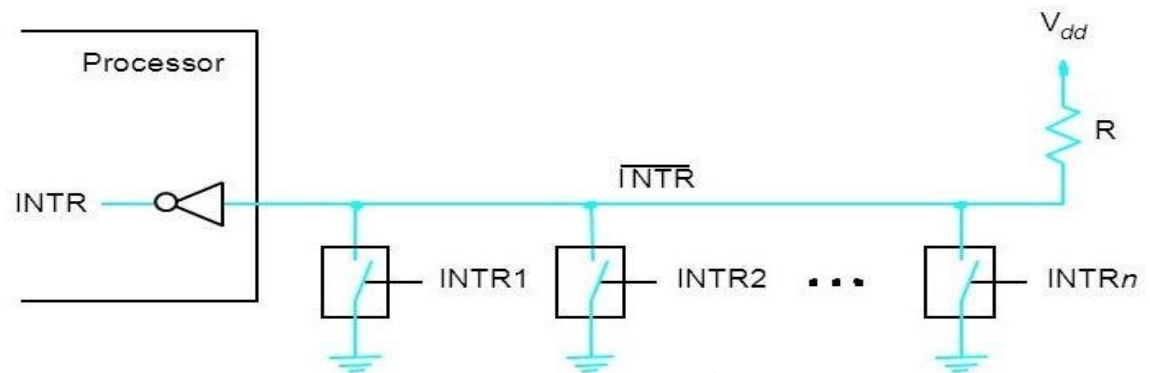


Interrupt Hardware:

- The external device (I/O device) request the processor by activating one bus line and this bus line is called as interrupt request line.
- The one end of this interrupt request line is connected to input power supply by means of pull up register is as shown in the fig.
- Another end of interrupt request line is connected to INTR (Interrupt request) signal of processor as shown in the fig.



Module-3 Input and Output Organization



- The I/O device is connected to interrupt request line by means of switch as shown in the fig.
- When all the switches are open the voltage drop on interrupt request line is equal to the VDD.
- This state is called as in-active state of the interrupt request line.
- The I/O device interrupts the processor by closing its switch.
 - When switch is closed the voltage drop on the interrupt request line is found to be zero. Therefore $INTR=0$ and $INTR=1$.
 - The signal on the interrupt request line is logical OR of requests from the several I/O devices. Therefore, $INTR=INTR1 + INTR2 + \dots + INTRn$



Enabling and Disabling the Interrupts:

- All computers fundamentally should be able to enable and disable interruptions as desired.
- The problem of infinite loop occurs due to successive interruptions of active INTR signals.
- There are 3 mechanisms to solve problem of infinite loop:
 - 1) Processor should ignore the interrupts until execution of first instruction of the ISR.

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- 2) Processor should automatically disable interrupts before starting the execution of the ISR.
- 3) Processor has a special INTR line for which the interrupt-handling circuit. Interrupt-circuit responds only to leading edge of signal. Such line is called edge-triggered.
- Sequence of events involved in handling an interrupt-request:
 - 1) The device raises an interrupt-request.
 - 2) The processor interrupts the program currently being executed.
 - 3) Interrupts are disabled by changing the control bits in the processor status register (PS).
 - 4) The device is informed that its request has been recognized.

In response, the device deactivates the interrupt-request signal.
 - 5) The action requested by the interrupt is performed by the interrupt-service routine.
 - 6) Interrupts are enabled and execution of the interrupted program is resumed.

Handling Multiple Devices:

While handling multiple devices, the issues concerned are:

- How can the processor recognize the device requesting an interrupt?
- How can the processor obtain the starting address of the appropriate ISR?
- Should a device be allowed to interrupt the processor while another interrupt is being serviced?
- How should 2 or more simultaneous interrupt-requests be handled?

Vectored Interrupt:

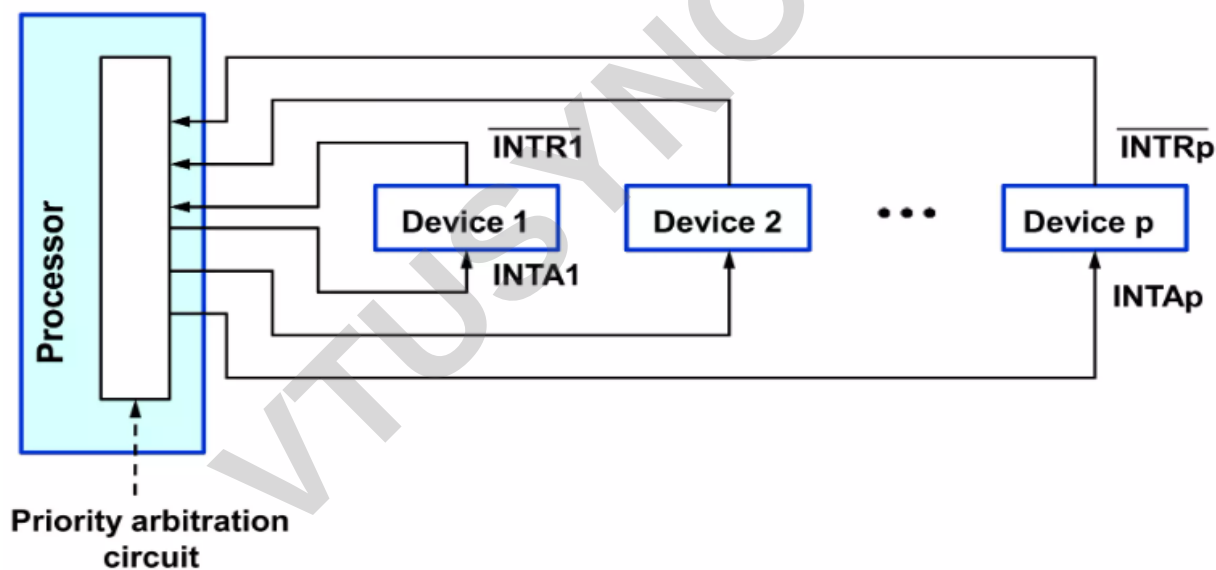
- A device requesting an interrupt identifies itself by sending a special-code to processor over bus.
- Then, the processor starts executing the ISR.
- The special-code indicates starting-address of ISR.
- The special-code length ranges from 4 to 8 bits.



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- The location pointed to by the interrupting-device is used to store the starting address to ISR.
- The starting address to ISR is called the **interrupt vector**.
- Processor
 - loads interrupt-vector into PC &
 - executes appropriate ISR.
- When processor is ready to receive interrupt-vector code, it activates INTA line.
- Then, I/O-device responds by sending its interrupt-vector code & turning off the INTR signal.
- The interrupt vector also includes a new value for the Processor Status Register.

Interrupt Nesting:



- A multiple-priority scheme is implemented by using separate INTR & INTA lines for each device
- Each INTR line is assigned a different priority-level as shown in Figure.
- Priority-level of processor is the priority of program that is currently being executed.
- Processor accepts interrupts only from devices that have higher-priority than its own.
- At the time of execution of ISR for some device, priority of processor is raised to that of the device.
- Thus, interrupts from devices at the same level of priority or lower are disabled.

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Privileged Instruction

- Processor's priority is encoded in a few bits of PS word. (PS = Processor-Status).
- Encoded-bits can be changed by **Privileged Instructions** that write into PS.
- Privileged-instructions can be executed only while processor is running in **Supervisor Mode**.
- Processor is in supervisor-mode only when executing operating-system routines.

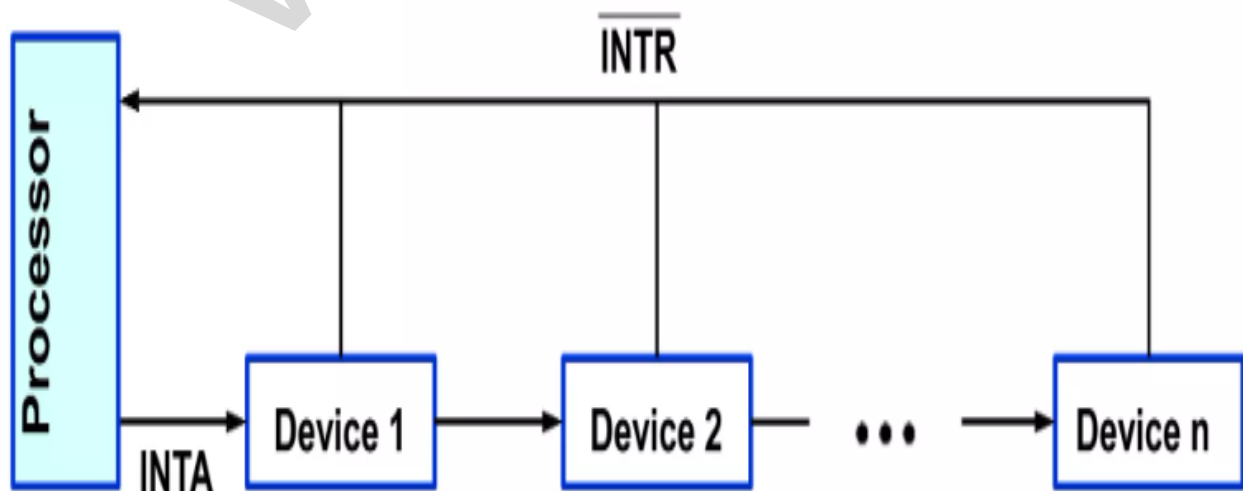
Privileged Exception

- User program cannot
 - Accidentally or intentionally change the priority of the processor &
 - disrupt the system-operation.
- An attempt to execute a privileged-instruction while in user-mode leads to a **Privileged Exception**.

Simultaneous Requests:

Daisy Chain

- The daisy chain with multiple priority levels is as shown in the figure.



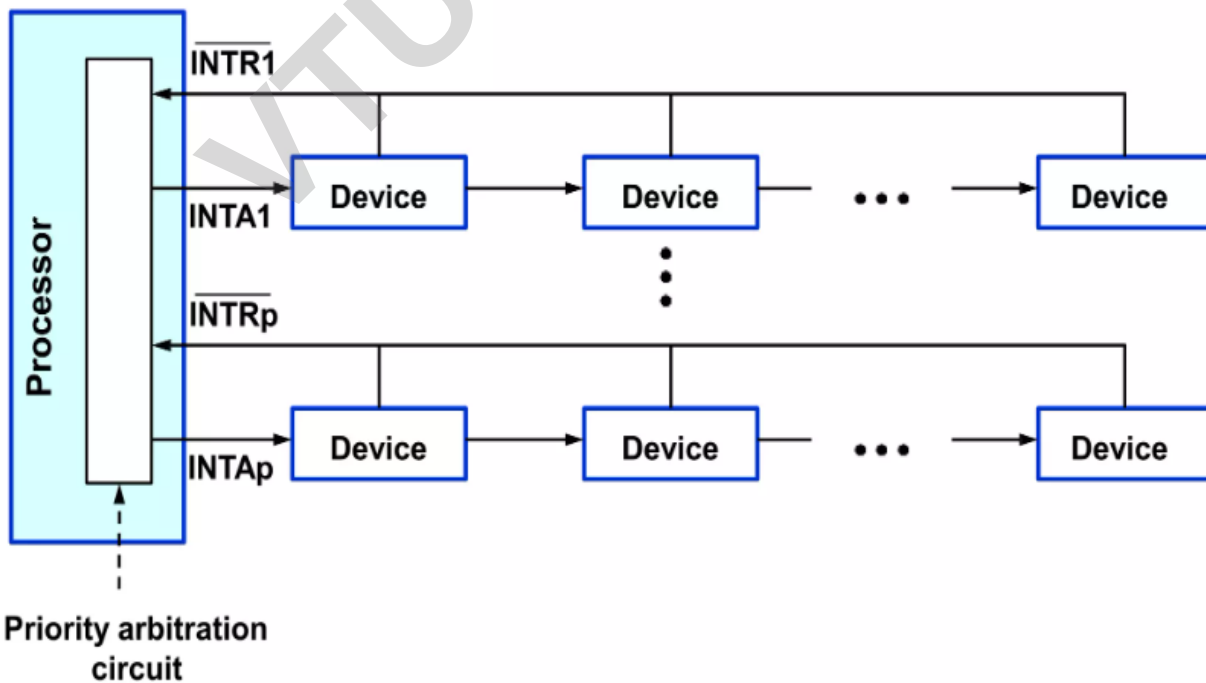
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- The interrupt request line INTR is common to all devices as shown in the fig.
- The interrupt acknowledge line is connected in a daisy fashion as shown in the figure.
- This signal propagates serially from one device to another device.
- The several devices raise an interrupt by activating INTR signal. In response to the signal, processor transfers its device by activating INTA signal.
- This signal is received by device 1. The device-1 blocks the propagation of INTA signal to device-2, when it needs processor service.
- The device-1 transfers the INTA signal to next device when it does not require the processor service.
- In daisy chain arrangement device-1 has the highest priority.
- **Advantage:** It requires fewer wires than the individual connections.



Arrangement of Priority Groups:

- In this technique, devices are organized in a group and each group is connected to the processor at a different priority level.
- Within a group devices are connected in a daisy chain fashion as shown in the figure.



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Direct Memory Address (DMA):

- It is the process of transferring the block of data at high speed in between main memory and external device (I/O devices) without continuous intervention of CPU is called as DMA.
- The DMA operation is performed by one control circuit and is part of the I/O interface.
- This control circuit is called DMA controller. Hence DMA transfer operation is performed by DMA controller.
- To initiate Directed data transfer between main memory and external devices DMA controller needs parameters from the CPU.
- These 3 Parameters are:
 - 1) Starting address of the memory block.
 - 2) No of words to be transferred.
 - 3) Type of operation (Read or Write).



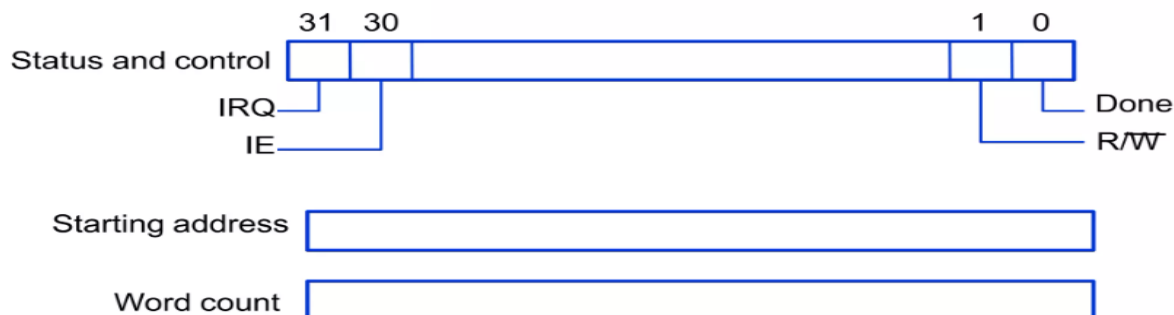
After receiving these 3 parameters from CPU, DMA controller establishes directed data transfer operation between main memory and external devices without the involvement of CPU.

• Register of DMA Controller:

It consists of 3 type of register:

Starting address register:

The format of starting address register is as shown in the fig. It is used to store the starting address of the memory block.



Word-Count register:

The format of word count register is as shown in fig. It is used to store the no of words

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to be transferred from main memory to external devices and vice versa.

Status and Controller register:

The format of status and controller register is as shown in fig.



a) DONE bit:

- The DMA controller sets this bit to 1 when it completes the direct data transfer between main memory and external devices.
- This information is informed to CPU by means of DONE bit.

b) R/W (Read or Write):

- This bit is used to differentiate between memory read or memory write operation.
- The R/W = 1 for read operation and
= 0 for write operation.
- When this bit is set to 1, DMA controller transfers the one block of data from external device to main memory.
- When this bit is set to 0, DMA controller transfers the one block of data from main memory to external device.

c) IE (Interrupt enable) bit:

- The DMA controller enables the interrupt enable bit after the completion of DMA operation

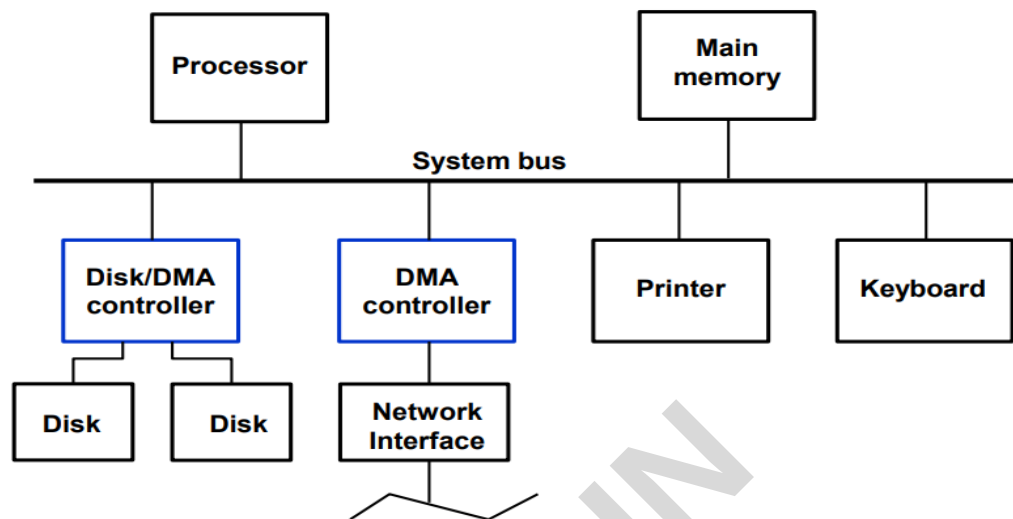
d) Interrupt request (IRQ):

- The DMA controller requests the CPU to transfer new block of data from source to destination by activating this bit.

The computer with DMA controller:

- The DMA controller connects two external devices namely disk 1 and disk 2 to system bus as shown in the above fig.
- The DMA controller also interconnects high speed network devices to system bus as shown in the above fig.
- Let us consider direct data transfer operation by means of DMA controller without the involvement of CPU in between main memory and disk 1 as indicated by dotted lines (in the fig.).

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- To establish direct data transfer operation between main memory and disk 1. DMA controller request the processor to obtain 3 parameters namely:
 - 1) Starting address of the memory block.
 - 2) No of words to be transferred.
 - 3) Type of operation (Read or Write).
- After receiving these 3 parameters from processor, DMA controller directly transfers block of data main memory and external devices (disk 1).
- This information is informed to CPU by setting respective bits in the status and controller register of DMA controller.
 - These are 2 types of request with respect to system bus
 - 1). CPU request.
 - 2). DMA request.
- Highest priority will be given to DMA request.
- Actually the CPU generates memory cycles to perform read and write operations.
 - The DMA controller steals memory cycles from the CPU to perform read and write operations. This approach is called as “**Cycle stealing**”.
- An exclusive option will be given for DMA controller to transfer block of data from external devices to main memory and from main memory to external devices. This technique is called as “**Burst mode of operation.**”